

SIMATS ENGINEERING

ASSESSMENT TOOL 1

ANALYTICAL PROBLEM SOLVING – SOLVED ASSIGNMENT

Course Name: Data Warehousing and Data Mining

Course Code: CSA16

Course Outcome: CO2 – Apply suitable Data Pre-processing techniques like Data Cleaning, Data Integration, Data Transformation and Data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50% **Total Questions:** 5 **Total Marks:** 50

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Q1. Covariance Analysis in Health Insurance

Question: The table shows annual premium increase rate (x) and claims per 100 policyholders (y). Using covariance, determine whether premium increase and claim frequency have a positive or inverse relationship. Before computing covariance, calculate the mean of x and y.

Step 1 – Means

$$\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 4.38$$

$$\bar{y} = (15 + 18 + 25 + 20) / 4 = 19.50$$

x	y	x - $\bar{x}$	y - $\bar{y}$	Product
3.2	15	-1.17	-4.50	5.29
4.0	18	-0.38	-1.50	0.56
5.5	25	1.12	5.50	6.19
4.8	20	0.42	0.50	0.21

Step 2 – Covariance

$$\text{Cov}(X, Y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / N$$

$$\Sigma \text{ products} = 12.25$$

$$\text{Cov}(X, Y) = 12.25 / 4 = 3.06$$

Interpretation: The covariance is positive. Therefore, higher premium increase rates tend to be associated with higher claim frequency in this sample. This indicates a positive relationship, although covariance alone does not establish causation.

Q2. Data Smoothing using Equal-Frequency Binning

Question: A hospital is analyzing systolic blood pressure readings. Apply local smoothing using equal-frequency (equi-depth) binning.

Step 1 – Equal-frequency partitioning: 27 observations are divided into 3 bins, giving 9 observations per bin.

Bin 1: 110, 112, 115, 115, 118, 120, 120, 122, 124

Mean = 117.33; Median = 118.00

Bin 2: 124, 126, 126, 126, 126, 130, 135, 135, 138

Mean = 129.56; Median = 126.00

Bin 3: 138, 138, 138, 140, 145, 150, 152, 160, 180

Mean = 149.00; Median = 145.00

Local smoothing by bin mean:

Bin	Original Values	Mean-Replaced Values
B1	110, 112, 115, 115, 118, 120, 120, 122, 124	117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33
B2	124, 126, 126, 126, 126, 130, 135, 135, 138	129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56
B3	138, 138, 138, 140, 145, 150, 152, 160, 180	149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00

Interpretation: Equal-frequency binning groups a similar number of records in each bin. Replacing observations with the bin mean smooths small fluctuations and reduces local noise. Extreme readings remain visible but their influence is reduced within their bin.

Q3. Equal-Width Binning for Risk Categorization

Question: An insurance company evaluates BMI values. Apply equal-width partitioning and replace values with the closest bin boundaries.

Step 1: Minimum = 18.5; Maximum = 39.5; Range = 21.0.

Bin width = $(39.5 - 18.5) / 3 = 7.00$.

Intervals:

Bin 1: 18.50 to 25.50

Bin 2: 25.50 to 32.50

Bin 3: 32.50 to 39.50

Original BMI	Nearest Boundary
18.5	18.50
19.2	18.50
21.8	18.50
24.9	25.50
25.5	25.50
26.2	25.50
26.4	25.50
27.8	25.50
29.2	32.50
30.2	32.50
30.4	32.50
31.9	32.50
32.4	32.50
33.1	32.50
33.6	32.50
34.7	32.50
38.2	39.50
39.5	39.50

Interpretation: Equal-width binning divides the numerical BMI range into equal intervals. Replacing values with the nearest boundary reduces small variations and creates a simplified representation suitable for analysis. The choice of boundary treatment should be documented because it affects the resulting values.

Q4. Data Reduction using Bin Medians

Question: A hospital administration is reviewing medical consumable costs. Use bin medians to discretize the sorted price records and reduce data complexity.

Sorted dataset: 120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 180, 190, 200, 210, 230, 300

Bin	Values	Median	Reduced Values
B1	120, 140, 145, 145, 148, 150, 150, 150, 155	148	148, 148, 148, 148, 148, 148, 148, 148, 148
B2	155, 155, 155, 160, 160, 165, 170, 170, 175	160	160, 160, 160, 160, 160, 160, 160, 160, 160
B3	175, 175, 175, 180, 190, 200, 210, 230, 300	190	190, 190, 190, 190, 190, 190, 190, 190, 190

Interpretation: Bin-median smoothing replaces every value in a bin with the median of that bin. This reduces the effect of local fluctuations and produces a compact representation of the original data. It is especially useful when the goal is to reduce noise while retaining the central tendency of each group.

Q5. Standardization of Patient Age Data

Question: Given ages 18, 22, 25, 42, 28, 43, 33, 35, 56, 28: (a) compute MAD; (b) compute z-score for the first four observations.

Step 1 – Mean age

Mean = $330 / 10 = 33.00$ years

Age	Age – Mean
18	15.00
22	11.00
25	8.00
42	9.00
28	5.00
43	10.00
33	0.00
35	2.00
56	23.00
28	5.00

Step 2 – Mean Absolute Deviation

MAD = $\sum |x_i - \bar{x}| / N = 88.00 / 10 = 8.80$ years

Step 3 – Standard deviation

Using population standard deviation: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 / N]} = 10.8351$

Step 4 – Z-scores for first four observations

Observation	Age	Z-score
1	18	-1.3844
2	22	-1.0152
3	25	-0.7383
4	42	0.8306

Interpretation: The z-score indicates how many standard deviations an observation lies above or below the mean. Ages 18 and 22 are below the mean, while ages 25 and 42 are respectively below and above the mean. Standardization puts the values on a common scale for analytical and machine-learning applications.

FINAL RESULTS SUMMARY

Question	Technique	Final Result / Interpretation
Q1	Covariance	Covariance = 3.06; positive relationship
Q2	Equal-frequency binning	27 records → 3 bins × 9; smoothing reduces local noise
Q3	Equal-width binning	Bin width = 7.00; 3 equal-range intervals
Q4	Bin median reduction	27 records → 3 bins × 9; each bin represented by its median
Q5	MAD + Z-score	MAD = 8.80; population $\sigma = 10.8351$; first four z-scores calculated

CONCLUSION

The assessment demonstrates how preprocessing techniques improve data quality and analytical readiness. Covariance helps identify relationships, binning techniques smooth and discretize numerical data, bin medians reduce noise, and z-score standardization places observations on a comparable scale. These techniques form an important preprocessing stage before Data Mining and Machine Learning.